

ITA0608 - Machine Learning

List of Lab Exercises

6. Write a program to implement Naïve Bayes algorithm in python and to display the results using confusion matrix and accuracy.

Aim

To implement the **Naïve Bayes algorithm** using the Iris dataset and evaluate its performance using the **confusion matrix** and **accuracy**.

Algorithm

1. Import the required libraries.
2. Load the Iris dataset.
3. Split the dataset into training and testing sets.
4. Create a Gaussian Naïve Bayes classifier.
5. Train the model using the training data.
6. Predict the class labels for the test data.
7. Display the confusion matrix.
8. Calculate and display the accuracy

Program Code

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import confusion_matrix, accuracy_score,
ConfusionMatrixDisplay

import matplotlib.pyplot as plt


iris = load_iris()

X = iris.data
```

```

y = iris.target

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.3, random_state=1)

model = GaussianNB()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

cm = confusion_matrix(y_test, y_pred)

print("Confusion Matrix:")

print(cm)

acc = accuracy_score(y_test, y_pred)

print("\nAccuracy:", acc)

disp = ConfusionMatrixDisplay(confusion_matrix=cm,

                               display_labels=iris.target_names)

disp.plot(cmap='Blues')

plt.title("Confusion Matrix")

plt.show()

```

Output

```

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score, ConfusionMatrixDisplay
import matplotlib.pyplot as plt

iris = load_iris()
X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1)

model = GaussianNB()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:")
print(cm)

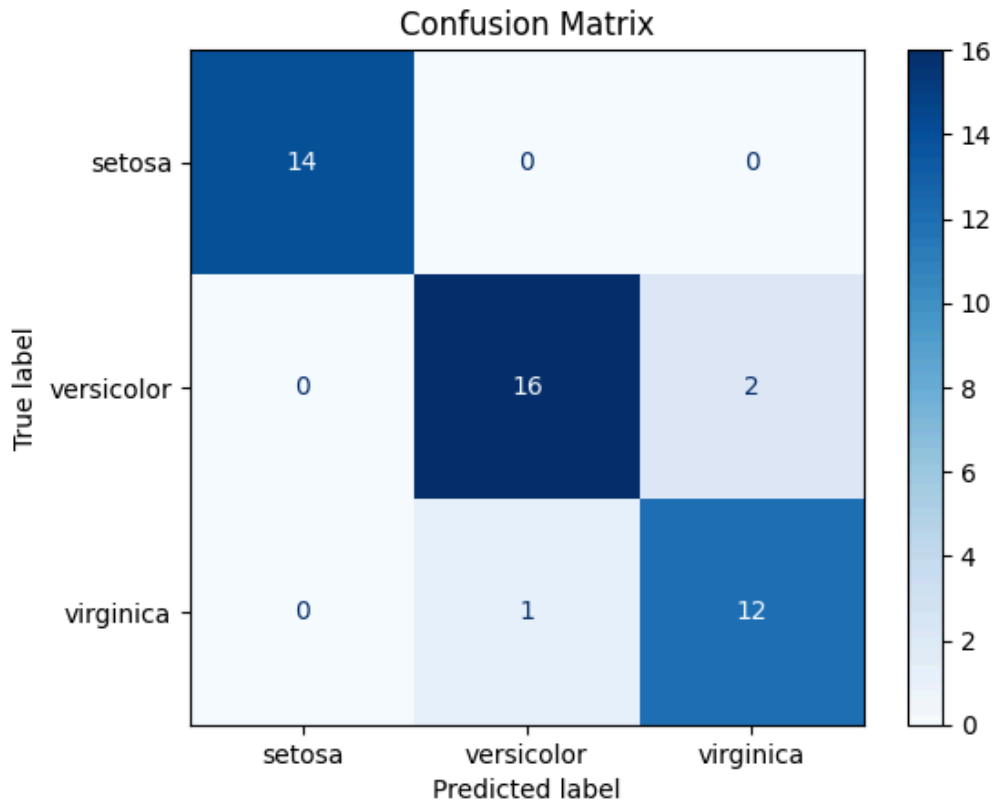
acc = accuracy_score(y_test, y_pred)
print("\nAccuracy:", acc)

disp = ConfusionMatrixDisplay(confusion_matrix=cm,
                              display_labels=iris.target_names)
disp.plot(cmap='Blues')
plt.title("Confusion Matrix")
plt.show()

```

```
*** Confusion Matrix:  
[[14  0  0]  
 [ 0 16  2]  
 [ 0  1 12]]
```

Accuracy: 0.9333333333333333



Result

The **Naïve Bayes classifier** was implemented successfully. The model classified the Iris dataset with high accuracy, and the confusion matrix showed the classification performance.